INCH POUND

MIL-C-3098/144B
23 December 1991
SUPERSEDING
MIL-C-3098/144A
19 March 1982

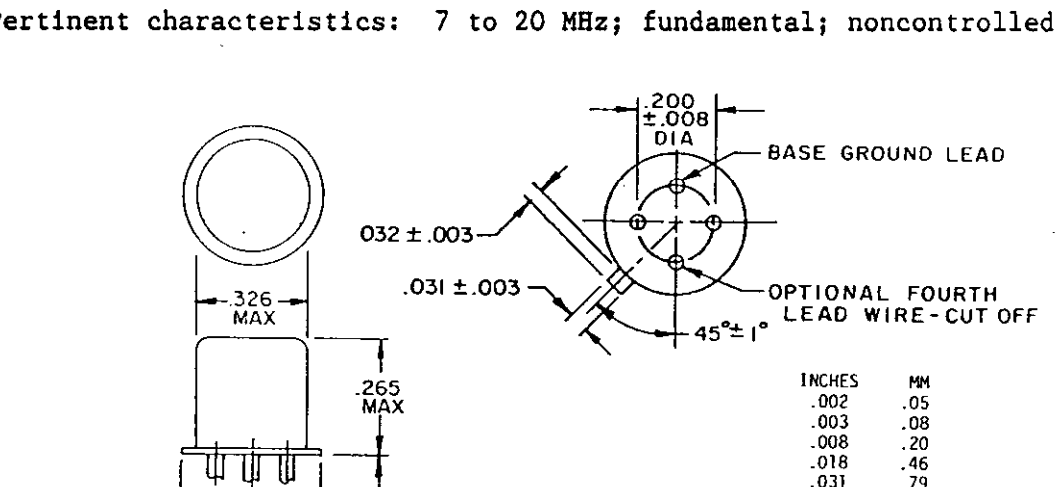
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR163/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 7 to 20 MHz; fundamental; noncontrolled; series.



INCHES	MM
.002	.05
.003	.08
.008	.20
.018	.46
.031	.79
.032	.81
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR163/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-35/U (MIL-H-10056/21).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 7.0 to 20 MHz, inclusive.

Frequency tolerance:
±40 ppm at -55°C to +105°C.
±20 ppm at -42°C to +90°C.

Equivalent resistance:
7.0 to 10.0 MHz = 30 ohms.
10.0+ to 20.o MHz = 20 ohms.

Mode of oscillator: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level:
7.0 to 10.0 MHz = 1.3 ±0.2 mW.
10.0+ to 20 MHz = 0.5 ±0.2 mW.

Capacitance, shunt: 7.0 pF, maximum.

Shock:
Frequency change permitted: ±5 ppm.
Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition D.
Frequency change permitted: ±5 ppm.
Equivalent resistance change permitted: ±10 percent.

Temperature cycling:
Frequency change permitted: ±3 ppm.
Equivalent resistance change permitted: ±10 percent.

Aging:
Frequency change permitted: ±3 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0660)

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